



Dynamics of Operational Efficiency in Credit Lending and Recovery of Stressed Assets: An Alternative Approach with Undesirable By-Products

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Accepted: 17 March 2024

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Abstract

Our study estimates the operational efficiency of Indian banks during 2009–2010 to 2017–2018, considering advances, and recovery of stressed assets as desirable outputs, while NPAs and slippages are undesirable by-products. Our first stage DEA analyses conclude that public sector banks have significant scope to improve their lending and recovery efficiency, while also reducing their stressed assets. Our analyses also reveal that banks are more efficient in managing credit risks for shorter-term loan portfolios and secured loans, while positive economic externalities are more prominent in determining the efficiency of banks in priority sector lending. We also observe that a competitive scenario within the banking sector, overall macroeconomic growth, and low cost of funds play vital roles in improving banks' operational efficiency and reducing credit risk. Our findings suggest that liquid assets might be more beneficial than illiquid collaterals in improving the recovery of stressed assets.

Keywords Financing Policy · Financial Risk and Risk Management · Banks' Ownership Structure · Stressed Assets · Slippage

JEL Classification D24 · G21 · G32 · G38 · L2

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Introduction

The Indian banking industry has undergone significant changes in the recent past, especially with mergers and acquisitions. Regulators and policymakers are concerned about the quality of assets and the inefficiency of recovering stressed assets in the current economic conditions. Our study focuses on improving recovery-led operational efficiency while managing stressed assets in Indian commercial banks. We examine the determinants of such performance, considering both public and private sector banks in a comparative framework. The low recovery rate of defaulted loan accounts is a matter of concern and requires policy deliberation to safeguard the interest of the Indian banking system and the growth of the economy. A decline in loan recovery not only mandates higher provisions and low profitability but also reduces a bank's lending capacity, which adversely affects the economy.

There has been extensive academic research measuring operational efficiency and productivity growth in the banking sector across different countries, including India. Additionally, there is a vast literature focused on the determinants of Non-Performing Assets (NPAs). It is important to note that measuring the efficiency of banking institutions helps to compare their relative performance to the best practice banks. Thus, measuring the Technical Efficiency Score (TES) of recovery of stressed assets, including NPAs and slippage, is crucial in the aftermath of a cyclical downturn when banks are dealing with a large volume of NPAs and low recovery.

There have been several official policy measures to improve the recovery channels of stressed bank assets, such as the Securitization and Reconstruction of Financial Assets and Enforcement of Security Interest (SARFESI) Act, 2002, Debt Recovery Tribunals, Lok Adalats, and the latest is the enactment of the Insolvency and Bankruptcy Code (IBC), 2016. These measures are welcome developments toward improving the credit environment and enhancing the credit-debtor relationship. However, their impact was not long-lasting (Misra et al. 2016).

In this context, our study aims to understand the factors that enable greater efficiency in the recovery of defaulted loans. Particularly, given the extensive mergers and acquisitions of public sector banks, this study sheds light on policy deliberation. Over time, with nationalization in 1969 and 1980, and later financial reforms since 1991, the banking system continues to be the strongest financial pillar of the Indian economy. An effective banking system has significant positive externalities and also has proved to be more instrumental in laying the inclusive growth path for the Indian economy. In fact, the operational inefficiency in managing the credit portfolio or credit risk of the bank might be proven to be detrimental to India's long-term economic growth. The low credit risk of a bank requires more attention towards two important factors: the overall *NPA* situation to improve and *slippage* to reduce. As for the treatment of the stressed assets of the Indian banks in the literature is concerned, Jayaraman and Srinivasan (2016) consider *GNPA* as an undesirable output, along with two other desirable outputs and four inputs, to study the *profit efficiency* of the Indian banks. On the other hand, Das et al. (2005) and Zaman and Bhandari (2020) consider *performing loans* (i.e., the amount of *loans* after due adjustment for

the volume of *non-performing loans*) to study *cost efficiency*¹ of the Indian banking sector. Although the study by Misra et al. (2016) and Bajaj et al. (2021) caters to the recovery of bad loans of the Indian banks, the focus of their study was to investigate the determining factors of such reduction of stressed assets. Also, both these studies have considered the *reduction in stressed assets*,² whereas the focus of our study is to examine the bank's operational efficiency in *cash recovery of stressed assets*. To the best of our knowledge, literature on estimating operational efficiencies of Indian banking and their determinants with due consideration for *stressed assets*, in general, and that concerning the *cash recovery* of such assets, in particular, is indeed scanty.

Given this backdrop, we examine TES of 21 public sector and 17 private sector banks during 2009–2010 through 2017–2018.³ We consider *four* alternative models to address four possible interlinked scenarios:

Scenario 1: To increase *advances*, while maintaining the current *GNPA* level;

Scenario 2: To increase *advances*, while maintaining the *net slippage*, defined as (*slippage* – *recovery*) at the current level;

Scenario 3: To simultaneously (proportional) increase both *advances* and *recovery* while maintaining the current *slippage* level; and

Scenario 4: To simultaneously (proportional) increase both *advances* and *recovery*, with parallel reduction of *slippage* in the same proportion.

Needless to note that, each of these models has its own appeal, considering the overall burden of stressed loans that the Indian banks face these days, possible channels of its improvement, and their day-to-day regular business activities. *Recovery* is considered in Models 3 and 4 as desirable output along with *Slippage* as undesirable output mostly from an operational outlook that there exists a one-to-one negative correspondence between these two outputs, given that both of these variables are highly linked with the country's overall macroeconomy. Given a better macroeconomic environment, recovery of stressed assets is likely to be high, while additional NPA (i.e., *slippage*) would be less. In short, our specific contribution is 1) bringing overall recovery aspects; and 2) treating undesirable output, for instance, stressed assets, into the analyses.

In the second stage, we explain such TES in terms of bank-specific and other macroeconomic and banking-sectoral characteristics in order to throw some light on the determining factors of such performance indicators of the Indian domestic commercial banks. To the best of our knowledge, our study is the first attempt which contribute to measuring the TES for credit risk of the Indian banks considering individual bank efficiency in managing *NPA* and *recovery* simultaneously.

Our major preliminary findings suggest that the performance of private sector banks in managing *recovery* is much better compared to public sector banks, although NPA shows a rising trend for both groups. Hence, the scope of

¹ To add further, Das et al. (2005) also study *profit* and *revenue efficiencies* for Indian banking sector.

² As per bank's annual report, reduction of NPA can have three segments: reduction from upgradation, reduction from write off, and reduction from cash recovery.

³ Due to large scale merger drives of the public sector banks, we keep aside the later periods from our analyses.

improvement in managing credit risk for public sector banks is more as compared to private sector banks. This is corroborated in our first-stage results using data envelopment analysis and even in each of the four alternative models explored. Most interestingly, we find that banks have vast scope to improve in terms of efficiency in dealing with the *recovery* even while *GNPA* and *slippage* are kept at their existing levels.

The rest of the paper is organized as follows. In “[Literature Review](#)”, we present a comprehensive review of the studies that have examined the NPA management and recovery of the banking sector. Also, we refer to a few studies which measure the performance of the Indian banking sector. In “[Analytical Background](#)”, we preclude the study with descriptive statistics, and “[Methodology and Database](#)” provides a brief background on the database and methodology followed in our study. In “[Empirical Findings](#)”, we summarize our findings of different models explored for measuring the technical efficiency scores of Indian commercial banks. Also, our findings on second-stage regression analyses have been discussed in this section, while “[Conclusion](#)” concludes with some policy implications. The appendix shows the results of *Technical Efficiencies* of the respective banks and the groups.

Literature Review

Estimation of efficiency score for managing credit risk of individual banks in terms of reduction in *NPA* and increase in *recovery* is critical to improve the overall performance of the banks. It is also important to examine the significant determinants of the operational efficiency of the banks. There are many studies that address the determinants of NPA or stressed assets, however, there exists merely one study that addresses the efficiency in dealing with the credit risk of the banks. Moreover, hardly any study refers to the determinants of efficiency in bank performance in controlling NPA and increasing the cash recovery of Indian commercial banks. It may be noted that most of the studies examining recovery rates pertain to advanced economies. By focusing on account-wise data, researchers have analyzed the recovery rate and its distribution. There is hardly any study that focuses on the emerging market economies, including India.

Asarnow and Edwards (1995) examine the US market scenario by delineating the components of Loss Given Default (LGD) such as write-offs, interest drag, cash interest collected, recoveries, and some other expense or income events. They test 831 defaulted loans of Citibank during 1970–1993 and report an average cumulative recovery rate of 65% by employing the discounted cash flow method. Another study based on the US market, namely Moody’s bank loan by Carty and Lieberman (1996), measures the recovery rate on a sample of 58 defaulted bank loans based on secondary market prices for the period 1989–1996 and reports an average recovery of 71%. In Latin America, Hurt and Felsovalyi (1998) investigated 1149 bank loan losses over the period 1970 to 1996. They report an average recovery rate of 68.2%. Interestingly, they also show that large loans exhibit lower recovery rates. More often so, when the large loans are not secured and are made to economic groups that are family-owned. Likewise, Dermine and Carvalho (2006) investigated LGD

characteristics for 374 corporate bank loans to small and medium-sized firms in Banco Commercial Portugues over the period 1995 to 2000 and reported an average recovery rate of 71%. Also, Khieu et al. (2012) observed that in the US, secured loans have higher recovery in comparison to unsecured loans.

A cross-country study by the Standard and Poor's Risk Solution Department (Franks et al. 2004) on loans to small and medium-sized enterprises reveals that, besides collateral, recovery rates also differ across countries where banks respond to different bankruptcy regimes and codes. On the other hand, the study on loan losses related to SME loans from 2001 to 2005 in the Eastern European Markets by Kosak and Poljsak (2010) concludes that the type of collateral and industrial sector, loan rating, size of the debt and loan maturity are important determinants of LGD. Similarly, Grunert and Weber (2009) observe a positive relationship between collateral and recovery rate for commercial lending to German companies.

Frye (2000) concludes that a high default period might make LGD more sensitive and the losses of the creditors to increase. Grunert and Weber (2009) suggest a negative relation between obligor creditworthiness and recovery rates. Also, distress in the situation of the industry itself might cause low recovery (Acharya et al. 2007). Caselli et al. (2008) assess the sensitivity of LGD to systematic risk in the Italian loan market and report that households are more sensitive to unemployment rates, default-to-loan ratio, and household consumption.

The probability of default (PD) is another important but sensitive variable in the context of the credit risk of the banks. However, the major constraint in using such data is the non-availability of the account-wise information in the public domain. In India, banks are extremely vigilant in maintaining high confidentiality of their clients' data. However, there are some studies that use the PD-LGD correlation model framework and conclude that the recovery is correlated with the firm's underlying asset process via both systematic factors and idiosyncratic shocks. PD-LGD correlation introduces additional variability into instrument value and portfolio value distributions, as viewed by Moody's in 2010. Altman et al. (2004) study the defaulted bonds' data for the period 1982–2000, which includes the relatively high default years of 1999 and 2000, and observe a negative correlation between the rates of default and recovery. Witzany (2011) studies default and recovery data on a retail portfolio of a large Czech bank from 2002 to 2008 and observes that the PD and LGD correlation is positive (viz., 0.0775), although the level of significance was low. However, too short a sample period may be a reason for such low-level significance.

On the other hand, there exists a plethora of studies in the context of measuring the efficiency score of Indian banks. To selectively elaborate on some of those, Das et al. (2005) the mathematical programming-based data envelopment analysis (DEA) method and observed that the Indian banks differ much in terms of revenue and profit efficiencies from 1997 to 2003. They also conclude that the *size* of a bank and its *ownership category* are important determining factors for average profit efficiency and (to some extent) revenue efficiency. Moreover, the median efficiency scores of Indian banks, in general, and of bigger banks, in particular, have improved during the post-reform period. Das and Ghosh (2006) examine a set of variables, for example, *bank size*, *ownership*, *capital adequacy ratio*, *non-performing loans*,

and *management quality* by employing three different models, viz., intermediation approach, value-added approach, and operating approach. They observe that medium-sized public sector banks perform reasonably well and are more likely to operate at higher levels of technical efficiency. Their empirical results also show that the more efficient banks, on average, have less non-performing loans.

Bhandari (2012) studies overall total factor productivity (TFP) improvement for 68 Indian commercial banks during 1998–1999 to 2006–2007. Using DEA method, this study examines all three constituent components of TFP growth, viz., technical change, technical efficiency changes, and scale (efficiency) change factor and suggests that the public sector banks, on an average, are better performers in a changing environment relative to private and foreign banks operating in India. Using a two stage method, i.e., DEA at the first stage, followed by panel data econometric regression at the second stage, Bhandari (2014) also examines 68 Indian commercial banks for the period 1998–1999 through 2006–2007. The results show that, technical efficiency is positively linked to size of a bank. Moreover, public sector banks are observed to perform much better than the private and foreign banks.

Profit efficiency of banks, factoring desirable and undesirable outputs, is first estimated by Jayaraman and Srinivasan (2016). They use Nerlovian profit indicator approach and decompose the profit inefficiency into technical and allocative inefficiencies. Their results using directional distance function reveal that, profit inefficiency of banks is primarily due to allocative inefficiency. In a recent study, Zaman and Bhandari (2020) examine cost efficiency of Indian banking using NPA-adjusted loans as an output. Since NPAs can be thought of as a by-products of a bank's conventional lending activities, it would have a direct impact on its performance.

Misra et al. (2016) have discussed several measures, which have been initiated to create enabling legal and regulatory environment to facilitate the recovery of NPAs of banks over the years. This study describes measures like 'know your customer' (KYC) norms, 'early warning signals', etc., and proposes curative measures (such as recovery channels) that are important to arrest the rise in NPAs. Moreover, using panel data regression on 71 public, private, and foreign banks' data from 2000–2001 to 2014–2015, they conclude that, other than a conducive macroeconomic environment, the collateral type and credit appraisal system of banks might be playing a more instrumental role in determining *reduction* of the stressed assets of Indian commercial banks. This study suggests that the alternative channels of the financial system also need to be strengthened. On the other hand, Bajaj et al. (2021) conclude that, between bank-specific and macroeconomic factors, the latter is more persuasive in the *reduction* of the stressed assets of public and private sector banks in India from 2003–2004 to 2017–2018. They have concluded that unemployment plays a significant role in determining banks' reduction of stressed assets.

So, a focused critical review of the literature states that there hardly exists any study that estimates the banks' operational efficiency considering *NPA* and *slippage* (which we treat as undesirable outputs) and *recovery* of stressed assets (which we treat as a desirable output) simultaneously. Although there is a spectrum of literature that examines the determinants of NPA, there hardly exists any study that investigates the causes behind the existing state-of-the-art operational efficiency of the recovery rate of Indian banks. We bridge this gap, especially, at a time when the

mergers and acquisitions of public sector banks have become a major concern subsequent to large NPAs and low recovery of Indian commercial banks. The next section throws light on our preliminary analyses where we hypothesize if there exists any visible difference in the trend of the performance of public and private sector banks in managing their credit risk.

Analytical Background

This section precludes our analysis, highlighting major differences in the trend of concerned important factors for public and private sector banks. Figure 1 shows that, although both the *GNPA* and *slippage* (or additional *NPA* of the respective year) follow the increasing trend for both public and private sector banks, there has been a sharp increase in the *NPA* accumulation for public sector banks since 2015, whereas the scenario for the private sector banks is not so gloomy. On the contrary, the *recovery* rate of the public sector banks sees a sharp declining trend over this period, however, although the private sector banks also experience (to some extent) a similar trend, its *recovery* declining tendency is not so sharp and even slightly improves in the last year. Hence, we conjecture that the performance of the private sector banks to manage both the *NPA* and *recovery* is better than the public sector banks. This differential trend is also visible in Table 1, which describes the trend in *GNPA* and *recovery* rates for the three largest banks under each of these two ownership categories. To be specific, the increasing *GNPA* trend is clearly visible in these three largest public sector banks (i.e., *SBI*, *PNB*, and *BOB*) with low recovery rates compared to not so gloomy scenario for the three largest private sector banks

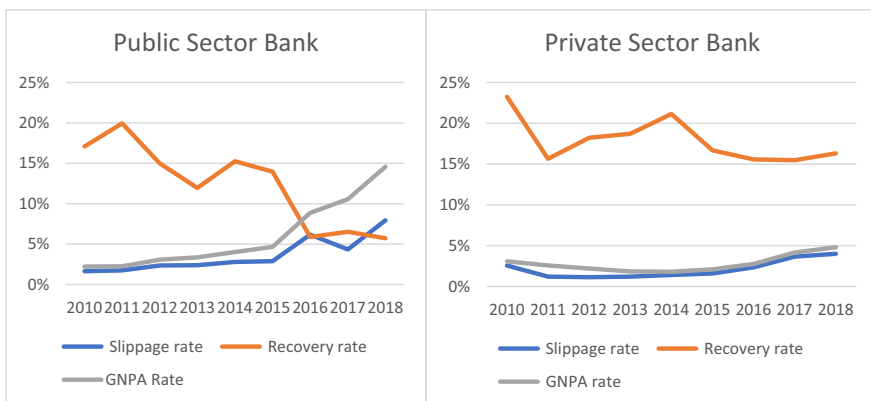


Fig. 1 Trend in slippage, GNPA and recovery rate of public and private sector banks (2009–2010 to 2017–2018). Source: Authors' Own Compilation using RBI Database. Slippage—Fresh Addition to NPAs during the year as a percentage of total standard advances at the beginning of the period. Strong credit appraisal and timely monitoring of loans are foundations to avoid slippages in the account. Recent crises suggest a negative relationship between rising slippages and recovery of the banks; Recovery—Cash recovery of the bank. This does not include the amount of write-off and/or upgradation; GNPA—Gross non-performing assets of the bank. Table 7 provides further details in this regard

Table 1 Comparison in GNPA rate and recovery rate among the three largest public and private sector banks during 2009–2010 to 2017–2018. Source: Authors' own calculation using RBI Database and Bank's Annual Report

Year	Bank of Baroda (BOB)		Punjab National Bank (PNB)		State Bank of India (SBI)	
	GNPA rate (%)	Recovery rate (%)	GNPA rate (%)	Recovery rate (%)	GNPA rate (%)	Recovery rate (%)
2010	2	16	2	30	3	11
2011	2	14	2	27	3	15
2012	2	13	3	19	4	10
2013	2	8	4	14	4	9
2014	3	11	5	14	4	13
2015	4	9	7	11	3	16
2016	10	4	13	8	5	4
2017	10	10	13	19	6	1
2018	12	8	18	5	11	2

Year	HDFC Bank		ICICI Bank		Axis Bank	
	GNPA rate (%)	Recovery rate (%)	GNPA rate (%)	Recovery rate (%)	GNPA rate (%)	Recovery rate (%)
2010	1	24	7	26	1	11
2011	1	9	6	14	1	16
2012	1	7	5	17	1	12
2013	1	31	3	12	1	11
2014	1	35	3	10	1	17
2015	1	41	4	8	1	9
2016	1	33	6	6	2	28
2017	1	29	9	11	6	9
2018	1	33	10	10	8	11

(i.e., *HDFC*, *ICICI* and *Axis*). Although GNPA is more in ICICI bank, it experiences a much higher parallel recovery.

Table 2 shows descriptive statistics of important business and financial health indicators of the Indian public and private sector banks during our study period. It may be noted that, from 2009–2010 to 2017–2018, the average *cash recovery* as a percentage of *GNPA* was only around 9% for the public sector banks, while that is around 17% for the private sector banks. This might be an early warning signal for the Indian banking sector to evaluate its credit risk portfolio. Moreover, we find public sector banks are larger contributors to average gross *advances* and average *GNPA* compared to their private sector counterparts. The average *slippage* for public sector banks is almost *seven times* higher compared to its own *recovery*, whereas the *former* is a little larger than four times the *latter* for the private sector banks. This might be indicative of the fact that, given the same macroeconomic environment, public sector banks are stumbling to manage their own credit risk. Moreover, the kurtosis values of *slippage* show that there exists a much fatter tail for the public

Table 2 Descriptive statistics for public and private sector banks during the sample period 2009–2010 to 2017–2018. Source: Authors' own calculation using RBI Database and Banks' Annual Report

	Slippage	GNPA	Gross advances	Cash recovery
<i>Public sector banks</i>				
Average (rupees crore)	9062.5	15,431.4	225,348.3	1332.4
Standard deviation (rupees crore)	14,778.0	23,846.7	317,674.8	1530.6
Coefficient of variation	163%	155%	141%	115%
Skewness	6.47	4.6	4.16	3.02
Kurtosis	59.77	32.5	18.48	12.35
<i>Private sector banks</i>				
Average (rupees crore)	1972.8	2477.5	79,131.6	421.9
Standard deviation (rupees crore)	4929.3	6626.0	124,087.6	793.3
Coefficient of variation	250%	267%	157%	188%
Skewness	4.90	5.33	2.55	3.70
Kurtosis	26.05	32.05	6.32	15.68

sector banks, although it is almost the same for both the groups for *GNPA*. This might be alarming for both public and private sector banks in terms of their exposure to high-value unexpected loss or credit risk. Interestingly, the kurtosis value of *recovery* for private sector banks is much higher than that of the public sector banks, which might protect the interests of private sector banks in recovering large amounts of stressed assets.

Tables 3, 4, 5 show the pairwise correlation structure for (a) major bank-specific variables; and (b) macroeconomic variables to have an impression of their likely implications on the performance of the Indian domestic commercial banks.

Table 3 shows that each of the *slippage*, *GNPA*, *recovery*, and *gross advances* has a significant positive pairwise correlation with others within the same ownership group. However, a negative and significant relationship is observed between public sector banks' *gross advances* with the private sector banks' *slippage* and *GNPA*. This triggers rather an important question of the account-wise lending preference is very different across the ownership patterns of the banks. To be specific, whether the lending accounts of the public sector banks cover the low-rating borrowers who are not fulfilling the credit appraisal criteria to borrow from private sector banks. However, given the scope of our study, we could not explore this particular aspect.

Table 4 shows that the pairwise correlations among bank-specific variables are mostly significant, which is negative for a *secured loan* (SecLoan) and *maturity*, indicating short-term loans are more secure than long-term loans. Moreover, the positive and significant correlation between the *priority sector lending* (PriSecL) norm and SecLoan is indicative of the fact that a bank's portfolio of priority sector lending is mostly secured. However, a negative and significant correlation indicates that PriSecL may not grow even if the *credit-to-deposit* (CD) ratio grows. On the other hand, a negative and significant correlation coefficient value indicates that an increase in bank *provisioning* (provision) might reduce the credit-to-deposit ratio. It means fewer business opportunities and less bank profitability. Moreover, a positive

Table 3 Correlation of credit variables of public and private sector banks comparison (2009–2010 to 2017–2018). Source: Authors' own calculation using RBI Database and Bank's Annual Report

	Private slip- page (Pr SL)	Private GNPA (Pr G)	Private gross advances (Pr GA)	Private recov- ery (Pr R)	Public slippage (Pu SL)	Public GNPA (Pu G)	Public gross advances (Pu GA)	Public recovery (Pu R)
Pr SL	1							
Pr G	0.94***	1						
Pr GA	0.77***	0.72***	1					
Pr R	0.89***	0.85***	0.79***	1				
Pu SL	-0.002	-0.04	0.03	0.04	1			
Pu G	0.04	-0.001	0.06	0.08	0.96***	1		
Pu GA	-0.13*	-0.15**	-0.1053	-0.1219	0.76***	0.79***	1	1
Pu R	-0.0428	-0.076	0.0083	-0.0239	0.59***	0.63***	0.62***	1

*, **, and *** indicate significant at respectively 10%, 5% and 1% level

Table 4 Correlation of bank specific variables during 2009–2010 to 2017–2018. Source: Authors' calculation using RBI database

	Maturity	SecLoan	PriSecL	CD	Provision	TL
Maturity	1					
SecLoan	– 0.43***	1				
PriSecL	– 0.34***	0.50***	1			
CD	0.35***	– 0.22***	– 0.41***	1		
Provision	– 0.06	0.16***	0.31***	– 0.35***	1	
TL	0.07	– 0.31***	– 0.33***	– 0.26***	0.22***	1

For description of the variables, see “[Methodology and Database](#)”

** and *** indicates significant at respectively 5% and 1% level

Table 5 Correlation of macroeconomic variables and HHI during 2009–2010 to 2017–2018. Source: Authors' own compilation using RBI Database

	LR	GDP growth	HHI
LR	1		
GDP Growth	– 0.57	1	
HHI	– 0.74**	0.59*	1

* and ** indicate significant at respectively 10% and 5% level

and significant coefficient implies that the *provision* also increases with the increase in the *total liability* (TL). Also, with the increase in *loan portfolio* (SecLoan and PriSecL), *provisioning* increases are evident from the positive and significant correlation coefficient.

In the Table 5, Herfindahl–Hirschman index (HHI) is calculated on the basis of *gross advances* of the banks to have a picture of the extent of competitive scenario prevails within the Indian banking sector. Positive and significant correlation indicates that the increase in GDP growth (due to increase in overall economic activities that perhaps also increases the demand for loan) induces market capturing competition amongst the banks to lend, however, smaller banks fail to effectively compete with the larger ones in that direction. As a result, banking industry becomes more concentrated, at least in terms of extending advances. On the other hand, the negative and significant coefficient indicates that the lending rate (LR) within the economy reduces due to this increased market capturing competitive pressure.

Methodology and Database

We use mathematical programming-based data envelopment analysis (DEA) methodology for our analyses in the first stage to obtain TES of banks under study. More specifically, we conceptualize the underlying production correspondence for a year by considering all the banks' information of that year alone, and the TES of a bank in a year is obtained by comparing it vis-à-vis that year's estimated production correspondence. Hence, we would have one TES for each bank in each year. This score

is used as a yardstick of the performance of a bank. We then regress these TESs in the second stage in terms of certain bank-specific and other macroeconomic factors, which are assumed to influence the performance of a bank. As for the second-stage regression methodology is concerned, we acknowledge that the separate cross-sectional regression for each year is preferred. It is advisable to follow so by virtue of the fact that the estimated technical efficiency scores (obtained at the first stage through DEA model) are not absolutely comparable over time, since these scores are relative measures—relative to the year-specific benchmark technology, where the benchmark itself is likely to change from one year to another. However, panel data regression is also used in the empirical literature in this regard (e.g., see Bhandari 2014; Chen et al. 2015; Samut and Cafri 2016; and others). Moreover, although Tobit regression is often proposed at this stage for the fact that the value of the dependent variable (i.e., the TES obtained at the first stage) is bounded between zero and unity by definition, it is however not obvious that Tobit is the only, or optimal, approach to modelling DEA scores. In fact, it is shown in the literature that the OLS may actually in many cases replace Tobit as a sufficient second-stage DEA model (e.g., see Hoff 2007; Banker and Natarajan 2008; and others). Hence, we follow (a) OLS for our second-stage regression analyses; and (b) both year-specific cross-sectional as well as longitudinal regression using the entire panel of nine years' data.⁴

DEA method, introduced by Charnes et al. (1978) (for the case when underlying production technology follows constant returns to scale (CRS) property) and further generalized by Banker et al. (1984) (to accommodate the case when underlying production technology follows more general variable returns to scale (VRS) property) requires no a priori parametric specification of the production frontier. On the basis of a sample of observed input–output data on a given set of producing units, it makes a few assumptions about production technology in order to obtain a production possibility set relevant for the observed units.⁵ Irrespective of the ownership, since each bank has some amount of non-performing assets (NPAs), we presume that such bad loan is a by-product of Indian banking business, which can not be avoided. Hence, bad loan is treated as an undesirable output.

Let us consider a production process that uses a vector of N inputs $x = (x_1, x_2, \dots, x_n, \dots, x_N) \in \mathfrak{R}_+^N$ to produce a vector of M desirable

⁴ Another strand of literature is to use bootstrap-based analyses to (a) obtain biased-corrected TES at the first stage; and (b) truncated regression for drawing consistence inference at the second stage, *a la* Simar and Wilson (2007). However, the question of bias at the conventional DEA estimation arises in the absence of true data generating process while using sample data. Nonetheless, we restrain ourselves from doing so, since we use almost the entire set of the banks under the two chosen groups (for which consistent data is available) and only a handful of them is left out. Hence, extent of bias, if any, is expected to be negligible.

⁵ The basic *assumptions* about the production technology that are made in this method are as follows: (a) All observed input–output bundles are feasible; (b) the production possibility set is *convex* implying that given a set of N feasible input–output bundles, *any* weighted average of these N input bundles can produce the same weighted average of the corresponding N output bundles and (c) any input or output is *freely disposable*. However, *free disposability* assumption needs to be adjusted if bad output(s) or some input(s) use of which beyond a threshold level may have detrimental effects on production process. Nonetheless, these assumptions enable one to construct a production possibility frontier on the basis of the observed inputs-output bundles of a given set of banks, following the DEA method.

outputs $y = (y_1, y_2, \dots, y_m, \dots, y_M) \in \mathfrak{R}_+^M$ and a vector of J undesirable outputs $b = (b_1, b_2, \dots, b_j, \dots, b_J) \in \mathfrak{R}_+^J$. The relationship between input and output is represented by the following output set:

$$P(x) = \{(y, b) : x \text{ can produce } (y, b)\}, x \in \mathfrak{R}_+^N. \quad (\text{a})$$

The output set is assumed to satisfy the following properties:

Null-Jointness It implies that production of strictly positive amount of desirable output must be accompanied by strictly positive amount of undesirable one. Formally,

$$\text{if } (y, b) \in P(x); b = 0 \Rightarrow y = 0 \quad (\text{b})$$

Weak Disposability It implies that desirable and undesirable outputs are jointly weakly disposable. Formally, if

$$(y, b) \in P(x) \text{ and } 0 \leq \theta \leq 1 \Rightarrow \theta(y, b) \in P(x) \quad (\text{c})$$

In other words, reduction in undesirable output is not possible without reducing the desirable one. So, free disposability of undesirable output may not be possible.

Strong Disposability Desirable output is strongly disposable, i.e., if

$$(y, b) \in P(x) \text{ and } y^0 \leq y, \text{ then } (y^0, b) \in P(x) \quad (\text{d})$$

Literally, although undesirable output cannot be reduced without reducing desirable output, the reverse is possible, i.e., the desirable output can be reduced without reducing the undesirable one. So, disposal of desirable and undesirable outputs are asymmetrically treated in our analyses. We measure TES of a bank in output-oriented way⁶ (i.e., expansion of desirable output, while keeping the undesirable output and usage of the inputs at their current levels). We assume VRS property for our underlying production technology. For that, we need to solve the following DEA linear programming problem (LPP), once for each bank. For instance, while evaluating k^{th} bank, the problem is:

Maximize β

⁶ One may also define an *input-oriented* technical efficiency. This is not done in the present study. See Coelli et al. (1998) and Ray (2004) for further details.

Table 6 Description of the alternative models used in the first stage DEA analyses. Source: Authors' compilation

	Model			
	TE 1	TE 2	TE 3	TE 4
Desirable output	Advances	Advances	Advances; recovery	Advances; recovery
Undesirable output	GNPA	Net slippage	Slippage	Slippage
Input	Total fixed assets; deposits; and operating expenses			

Models TE 1, TE 2, and TE 3 are based on LPP (1), while Model TE 4 is based on LPP (2); $M=1$ in Models TE 1 and TE 2 and $M=2$ in Models TE 3 and TE 4; $J=1$ and $N=3$ for all the four Models

We follow RBI definition for each of the variables; net slippage = slippage – recovery

Subject to

$$\begin{aligned}
 & \sum_{i=1}^P \lambda_i y_i^m \geq \beta y_k^m \quad \forall m = 1, 2, \dots, M; \quad (\text{i}) \\
 & \sum_{i=1}^P \lambda_i b_i^j = b_k^j \quad \forall j = 1, 2, \dots, J; \quad (\text{ii}) \\
 & \sum_{i=1}^P \lambda_i x_i^n \leq x_k^n \quad \forall n = 1, 2, \dots, N; \quad (\text{iii}) \\
 & \sum_{i=1}^P \lambda_i = 1 \quad \forall i = 1, 2, \dots, P \quad (\text{iv}) \\
 & \lambda_i \geq 0 \quad \forall i = 1, 2, \dots, P \quad (\text{v})
 \end{aligned} \tag{1}$$

where P is the number of banks in a year and TES of the k^{th} bank is given by $(1/\beta_k^*)$ where β_k^* is the optimal solution of the problem (1) above. Alternatively, we have also used directional (technology) distance function to represent the technology. On the basis of Luenberger's (1992) benefit function (see Chambers et al. 1996; Färe and Grosskopf 2000; and Färe et al. 2005), directional distance function, an extension of Shephard's input and output distance function, provides a platform for representing the joint production of *desirable* and *undesirables*. In the presence of undesirable output, if firm's objective is to simultaneously expand the desirable output and reduce the undesirable one by same proportion without increasing its input use, the directional technology distance function becomes

$$\overrightarrow{D}_T(x, y, b; 0, y, -b) = \sup [\beta : [(1 + \beta)y, (1 - \beta)b] \in P(x)] \tag{e}$$

The directional distance function is obtained by solving the maximization problem as follows:

Maximize β

Subject to

$$\begin{aligned}
 & \left\{ \sum_{i=1}^P \lambda_i y_i^m - \beta y_k^m \right\} \geq y_k^m \quad \forall m = 1, 2, \dots, M; \text{ (i)} \\
 & \left\{ \sum_{i=1}^P \lambda_i b_i^j + \beta b_k^j \right\} = b_k^j \quad \forall j = 1, 2, \dots, J \text{ (ii)} \\
 & \sum_{i=1}^P \lambda_i x_i^n \leq x_k^n \quad \forall n = 1, 2, \dots, N \text{ (iii)} \\
 & \sum_{i=1}^P \lambda_i = 1 \quad \forall i = 1, 2, \dots, P \text{ (iv)} \\
 & \lambda_i \geq 0 \quad \forall i = 1, 2, \dots, P \text{ (v)}
 \end{aligned} \tag{2}$$

and TES of the k^{th} bank is given by $\{1 / (1 + \beta_k^{**})\}$ where β_k^{**} is the optimal solution of the problem (2) above. Strong disposability of desirable output and weak disposability of undesirable output are imposed through the constraints (i) and (ii) respectively in each of the problems (1) and (2). We have followed four alternative models as summarized in the Table 6 below.

In the second stage of our analyses, we employ both cross-sectional and static panel regressions. We use the estimated TES of all four models as the dependent variable and estimate cross-sectional regression with bank-specific variables and an ownership dummy. Similarly, we estimate panel regression under four different scenarios: (i) considering the bank-specific variables alone; (ii) considering the bank-specific variables, along with banking industry concentration index HHI and macroeconomic variable; (iii) considering the bank-specific variables and interest rate channel; and (iv) considering the bank-specific variables and ownership dummy. Table 7 represents the description of the variables used in our second-stage regression analyses.

Except for the recovery data and that of the macroeconomic variables like GDP growth and LR, we use Reserve Bank of India (RBI) database of *Statistical Tables Relating to Banks in India (STRBI)*. We have extracted recovery data from the *Annual Reports* of respective banks, which provides the data on cash recovery. In the RBI database, we have data for opening GNPA of a particular year, addition during the year, reduction during the year, write-off during the year, and closing GNPA of that year (*Refer, STRBI, Table 6: Movement of Non-Performing Assets (NPAs) of Scheduled Commercial Banks*). In annual report of the bank, we get the data of reduction as the sum of upgradation, cash recovery and write-off. We have considered cash recovery during the year. This data is available consistently in the banks' annual reports from 2009–2010 onward. Information on GDP growth and LR are extracted from the world development indicator of the World Bank.

We first start with cross-sectional analysis to capture the impact of the significant determinants of the TES of the banks for each of the sample years. The equation for cross-sectional regression is represented by Eq. (3) below.

Table 7 Variable descriptions. Source: Authors' own compilation

Variable	Description
Net Interest Income ratio (NII)	Ratio of interest income minus interest expenses to total assets. It is an indicator of financial stability. A decrease in spread prompt the bank to compromise on quality of borrowers
Maturity	Ratio of share of term loans to total loans. This ratio is an indicator of longer term loan contract to fund medium to long-term projects, which reflect better relationship between banks and borrowers
Secured loan (SecLoan)	Proportion of secured loans to total loans. This ratio indicates how much collateral cushion the bank have against the loan amount and reflects the bank's approach towards risk management
Priority sector loan (PriSecL)	Priority sector loan to total loan. This ratio is taken in order to account for the argument that the priority sector loans are responsible for most number of defaults in banks in India
Credit–deposits ratio (CD)	It reflects credit orientation of the bank to earn more money through deposits. This variable is considered in order to capture aggressiveness of the banks' lending activity, which can lead to NPAs. The higher the ratio, the higher the loan-assets created from deposits
Provisioning (Provision)	Provision for NPAs/total loans: higher provision appears to reduce the level of net NPAs (doubtful and loss assets) and reduces bank's profitability. Higher provisioning percentage hints to improve recovery efforts, to free the blocked funds in alternative uses
Total liabilities (TL)	Total liabilities to income ratio. Deposit having the largest share in the total liability of the bank, also a key contributor to the business of the bank. (Annex 2, Banking Stability Map and Indicator)
Slippage	Fresh Addition to NPAs during the year as a percentage of total standard advances at the beginning of the period. Strong credit appraisal and timely monitoring of loans is a foundation to avoid slippages in the account. Recent crisis suggest negative relationship between rising slippages and recovery of the banks
Recovery	The cash recovery of the bank. This does not include the amount of write off and/or upgradation
GNPA	Gross non-performing assets of the bank
Operating expenses	It is an indicator of banks' usual day-to-day operations and other business-related activities in carrying out due diligence in application, credit deployment, monitoring and recovery of loans
Herfindahl–Hirschman index (HHI)	This is calculated by squaring the market share of each bank gross advances and then summing the resulting numbers. This is an indicator of market competition
GDP growth (GDP)	GDP growth rate is considers as a very importance macro-economic variable to judge default and recovery risk of the bank. When fundamental are good, default risk is less and recovery is more
Lending rate (LR)	Lending rate in India at which banks lend to its most creditworthy customer

Abbreviation, if any, is shown in the parenthesis

Banking Stability Map and Indicator—CAMELS rating (see the link <https://rbidocs.rbi.org.in/rdocs/PublicationReport/Pdfs/9FSRDECANEX21FA6045AFD51466D877FA56DFEB448F1.pdf>)

Table 8 Correlation of technical efficiency scores of the banks during 2009–2010 to 2017–2018. Source: Authors' own estimation using RBI Database and Annual Report of Individual Banks

	TE 1	TE 2	TE 3	TE 4
TE 1	1			
TE 2	0.80***	1		
TE 3	0.64***	0.78***	1	
TE 4	0.62***	0.74***	0.96***	1

** and ***represent significant at respectively 5% and 1% level

$$TES_{ji} = \beta_1(Maturity_i) + \beta_2(SecLoan_i) + \beta_3(PriSecL_i) + \beta_4(CD_i) + \beta_5(Provision_i) + \beta_6(TL_i) + \beta_7D + \alpha + u_i \quad (3)$$

In Eq. (3) we have used an ownership dummy defined as $D=0$ for all public sector banks, and $D=1$ for all private sector banks to examine the distinctive implication of ownership on TES, if any, and u_i is the idiosyncratic error term. Also, while 'i' symbolizes cross-sectional analysis, $j=1, 2, 3$, and 4 to specify four different models (as specified in Table 6) that we have explored.

For panel data regression analysis, we have considered two important channels, for example, *macroeconomic factors* and *concentration index* for lending, in addition to *bank-specific channels* and *ownership patterns of banks* which we have also used in the cross-sectional analysis. Under static panel regression analyses, we use both fixed effects and random effects models. The fixed effects model controls for all time-invariant differences between the individuals. It enables one to study the causes of changes within an entity and a time-invariant characteristic cannot cause such a change, because it is *fixed* for each individual. Our analysis of *bank-specific factors* is represented in *Panel Estimation-I*.

Panel Estimation-I

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(CD_{it}) + \beta_5(Provision_{it}) + \beta_6(TL_{it}) + \alpha_i + \varepsilon_{it} \quad (4)$$

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(CD_{it}) + \beta_5(Provision_{it}) + \beta_6(TL_{it}) + \alpha + u_{it} + \varepsilon_{it} \quad (5)$$

Here, in the specification of the Eq. (4) with α_i is the *fixed* unknown intercept for each entity. The rationale behind the random effects model is that, unlike the fixed effects model, the variation across entities is assumed to be random and uncorrelated with the predictor or the independent variables included in the model. In Eq. (5) *between entity error* is represented by u_{it} and *within entity error* is noted as ε_{it} .

Similarly, the *Panel Estimation-II* examines Eqs. (6) and (7) with *bank specific factors*, *macroeconomic factor* and *concentration index* for lending.

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(CD_{it}) + \beta_5(Provision_{it}) + \beta_6(HHI_i) + \beta_7(GDP_i) + \alpha_i + \varepsilon_{it} \quad (6)$$

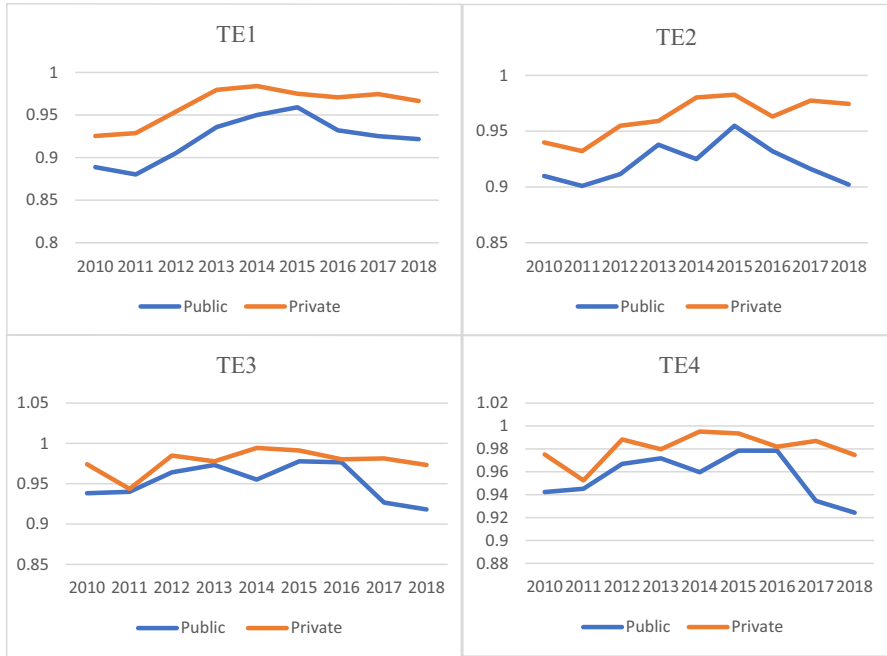


Fig. 2 Ownership-wise performance of Indian commercial banks. Source: Authors' Own estimation. Comparative performance as per the ownership pattern

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(CD_{it}) + \beta_5(Provision_{it}) + \beta_6(HHI_i) + \beta_7(GDP_i) + \alpha + u_{it} + \varepsilon_{it} \quad (7)$$

Panel Estimation-III examines Eqs. (8) and (9) capturing *bank specific factors* and *interest rate channel*.

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(CD_{it}) + \beta_5(Provision_{it}) + \beta_6(LR_i) + \alpha_i + \varepsilon_{it} \quad (8)$$

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(CD_{it}) + \beta_5(Provision_{it}) + \beta_6(LR_i) + \alpha + u_{it} + \varepsilon_{it} \quad (9)$$

Lastly, *Panel Estimation – IV* examines Eq. (10) which deals with the *bank specific factors* and the *ownership pattern of the banks*.

$$TES_{jit} = \beta_1(Maturity_{it}) + \beta_2(SecLoan_{it}) + \beta_3(PriSecL_{it}) + \beta_4(Provision_{it}) + \beta_5(D) + \alpha + u_{it} + \varepsilon_{it} \quad (10)$$

In all the above equations, j stands for efficiency scores derived from four different DEA models summarized in the Tables 6 and 8 and hence $j = 1, \dots, 4$. As usual,

i stands for cross sectional unit and varies from 1 to 38 (incorporating 21 public sector banks and 17 private sector banks) and t stands for time series unit and varies from 2009–2010 to 2017–2018,⁷ indexed respectively by 1 to 9 for all the equations.

Empirical Findings

First Stage DEA Analyses

We summarize the findings of our (first stage) DEA analyses in this section. Table 8 reports the pairwise correlation coefficients of estimated TES for each of the alternative models we have explored: TE 1 through TE 4. It shows that our results are robust throughout, since TES across all the pairs are positive and significant. Moreover, very high value of the correlation coefficient for the pairs of TE 1 and TE 2 (i.e., 80%), and TE 3 and TE 4 (i.e., 96%) are quite understandable from the similarity of the output vector for each of these pairs. We also plot average performance scenario of the group of publicly and privately owned domestic banks in Fig. 2 and the counterpart statistics are reported in the Table 18 (in the Appendix). It shows that, private sector banks as a group outperform those under public ownership. Moreover, although the two groups face similar tendencies of performances over time, that of the public sector group experiences a nosedive in the last two years in all the four alternative model specifications. In the banking performance literature in India, banks' efforts towards *recovery* of loans or reducing *slippage* or a combination of both is largely ignored. We explore that direction, considering *recovery* as one of the most significant factors and *advances* as traditional desirable outputs (in TE 3 and TE 4). And in all of these alternative cases, privately owned banks' overall performance is visibly quite better as compared to publicly owned banks. To be specific, the average TES of the public (private) sector banks as a group for our entire sample period are 92.2% (96.2%), 92.1% (96.3%), 95.2% (97.8%), and 95.6% (98.1%) respectively for the models TE 1 through TE 4. In other words, according to the TE 1, an average public (private) sector bank has an opportunity to expand its *advances* by 8.5% (4.0%)⁸ further while keeping its *GNPA* and three inputs (viz. *total fixed assets*; *deposits*; and *operating expenses*) at their current levels. As per TE 2, an average public (private) sector bank has an opportunity to expand its *advances* by 8.6% (3.9%) further while keeping its *net slippage* and three inputs at their current levels. Likewise, according to TE 3, an average public (private) sector bank has a scope to simultaneously expand its *advances* and *recovery* by 5.0% (2.3%) further while keeping its *slippage* and three inputs at their current levels. Finally for TE 4, an average public (private) sector bank has an opportunity to simultaneously expand its *advances* and *recovery* and reduce *slippage* by 4.7% (2.0%) further without increasing its usage of three inputs. Further details

⁷ For extensive mergers and acquisitions of public sector banks in India in the year 2018, 2019 and 2020, we exclude these years from our analyses. This helps us to keep the standard of the data without losing much of information.

⁸ To clarify further, TES=92.2% implies that there exists a further $\{[(1/0.922) - 1] \times 100\}$ % of output expansion opportunities.

on individual bank-wise average TES over our entire sample period is reported in the Table 16 in the Appendix. Table 17 is simply Table 16's counterpart, interpreting TES in terms of further possibility of (proportional) expansion of desirable output(s) (in models TE 1 to 3), and that along with contraction of undesirable output (in TE 4). As for the individual banks are concerned, State Bank of India & its associates, IDBI Bank, Corporation Bank, and Canara Bank are some of the best performing public sector banks, while HDFC Bank, ICICI Bank, Karnataka Bank, Nainital Bank, Ratnakar bank, and Yes Bank are so within the private sector group. On the other hand, United Bank of India, Dena Bank, Central Bank of India, and Allahabad Bank are some of the worst-performing banks under public ownership, while Jammu & Kashmir Bank and Tamilnad Mercantile Bank are relatively poor performers under the private ownership category.

Figure 2 presents a comparative framework of technical efficiency of banks on the basis of ownership pattern. Overall, we can conclude that private sector banks have been outperforming the public sector banks in managing credit risk with a widening gap over time. Especially, the sharp drifts in TE 3 and TE 4 models reveal that public sector banks have substantial scope to improve their technical efficiencies in increasing *advances* and *recovery* of stressed assets without increasing the *slippage* (TE 3) or simultaneously reducing the accumulation of *bad loans* (TE 4). Moreover, TE 1 (TE 2) reveals that given the current level of *GNPA* (*net slippage*) the technical efficiency of private sector banks in providing *advances* is notably higher compared to public sector banks.

Second Stage Econometric Analysis

Our estimated cross sectional regression results (reported in the Table 9) throw thorough light on the important factors determining technical efficiencies of Indian commercial banks during 2009–2010 to 2017–2018. Overall, we observe bank's technical efficiency increases with the increase in credit-deposit ratio. Moreover, banks are more efficient in better handling short-term loans than the long-term loans. As expected, with an increase in the share of secured loan to total loan, a bank's operational efficiency in dealing with the credit risk increases. It is also observed that the priority sector lending norm clearly has negative implication on banks' operational efficiency, however, which can be justified with its overall contribution in social development. Hence, we conclude that banks are able to handle the credit risk more efficiently for short term loan and there is further scope of improvement for the long-term loan management.

Interestingly, we find that the private sector bank is more efficient in handling the advances and recovery compared to public sector banks, with an exception for 2018. Also, a pooled regression result with a time dummy reveals that banks' performance in managing bad loans is exceptionally bad in 2011, however, the same is not true for recent years (Table 10). In 2011 and 2017, unexpectedly low TES are observed in efficiency estimation (Table 18) where one or few outlying observations pull up the entire technology frontier. However, our econometric estimation reported in Table 10 shows the break in 2011 is statistically significant, while the break in 2017

Table 9 Year wise cross section regression analysis to explain DEA technical efficiency scores. Source: Authors' own estimation

	2018			
	TE 1	TE2	TE3	TE4
Maturity	− 0.003**	− 0.002**	− 0.002**	− 0.002**
SecLoan	0.005**	0.006***	0.004*	0.004*
PriSecL	− 0.006**	− 0.007***	− 0.006**	− 0.005**
CD	0.007***	0.008***	0.007***	0.006***
Provision	− 0.004	− 0.008	− 0.004	− 0.003
TL	− 0.022	− 0.023	− 0.04**	− 0.04**
D	− 0.12**	− 0.11**	− 0.12**	− 0.11**
_cons	0.68	0.54	0.98**	0.996**
Adj R-square	51%	58%	49%	47%

	2017			
	TE 1	TE 2	TE 3	TE 4
Maturity	− 0.003**	− 0.002*	− 0.002*	− 0.001
SecLoan	0.002	0.003	0.002	0.002
PriSecL	− 0.003	− 0.003	− 0.003	− 0.003
CD	0.005**	0.006	0.005**	0.004**
Provision	− 0.003	− 0.009	− 0.011	− 0.012
TL	0.0006	0.013	0.012	0.013
D	− 0.013	0.004	− 0.001	0.005
_cons	0.71	0.36	0.51	0.542
Adj Rsquare	34%	42%	42%	43%

	2016		2015		2014			
	TE 3	TE 4	TE 3	TE 4	TE 1	TE 2	TE 3	TE 4
Maturity	− 0.0006	− 0.0005	.0001**	− 0.001	− 0.00003	− 0.0003	− 0.0002	− 0.0002
SecLoan	0.003**	0.003**	0.01	− 0.001	− 0.0009	− 0.0002	− 0.0009	− 0.001
PriSecL	− 0.004**	− 0.004**	.0003**	0.001	− 0.0006	− 0.0009	0.0010	0.001
CD	− 0.0001	− 0.00016	− 0.009	0.003**	0.0008	0.0011	0.0012	0.001
Provision	0.002	0.000907	0.12	0.027	0.038*	0.044*	0.044**	0.037*
TL	− 0.021*	− 0.02*	0.026	0.018	0.022*	0.025**	0.022**	0.02**
D	− 0.005	− 0.005	0.245	0.05**	0.091***	0.12***	0.09***	0.08***
_cons	1.12***	1.13***	.071*	0.67**	0.6990***	0.56**	0.62***	0.68***
Adj Rsquare	8%	8%	12%	18%	14%	29%	19%	17%

Best fit model	2013		2012		2011				2010
	TE 1	TE 4	TE 1	TE 4	TE 1	TE 2	TE 3	TE 4	TE 1
Maturity	− 0.0010	− 0.0007	− 0.0003	− 0.001*	− 0.0007	− 0.002**	− 0.001*	− 0.0006	− 0.0008
SecLoan	0.0002	− 0.001	− 0.002	− 0.001	− 0.0009	− 0.002	− 0.002	− 0.0012	− 0.0010
PriSecL	− 0.0003	0.002	0.002	− 0.0004	− 0.0032	− 0.008*	− 0.005	− 0.0034	− 0.0012
CD	0.004***	0.005***	0.004**	0.001	0.0056**	0.004**	0.003**	0.002*	0.007**
Provision	− 0.035	0.04	− 0.076	− 0.024	0.032	− 0.008	0.017	0.011	0.016

Table 9 (continued)

Best fit model	2013		2012		2011				2010
	TE 1	TE 4	TE 1	TE 4	TE 1	TE 2	TE 3	TE 4	TE 1
TL	0.017	0.03**	− 0.003	0.009	− 0.017	− 0.018	− 0.005	0.002	− 0.011
D	0.033	0.06**	0.001	0.021	0.029	0.021	0.010	0.017	− 0.001
_cons	0.51*	0.31	0.80**	0.98***	0.88**	1.31***	1.12***	0.98***	0.723
Adj R-square	25%	19%	15%	10%	20%	28%	18%	13%	23%

2016 onwards we have presented only the best fitted model selected on the basis of the Adjusted R-square value for the respective years. Variable description is reported in Table 7

*, **, and *** represent significant at respectively 10%, 5% and 1% level

Table 10 Pooled regression results with time dummy. Source: Authors' own estimation

	TE 1	TE 2	TE 3	TE 4
Maturity	− 0.0007**	− 0.0009***	− 0.0008***	− 0.0006***
SecLoan	0.0009	0.0002	− 0.00004	0.00005
PriSecL	− 0.0013*	− 0.001*	− 0.002***	− 0.0014***
CD	0.0041***	0.004***	0.0025***	0.0021***
TL	− 0.0018	0.002	− 0.0032	− 0.0033
D11	− 0.035***	− 0.029**	− 0.024**	− 0.019**
D17	0.0190	0.016	− 0.003	− 0.0011
_cons	0.67***	0.70***	0.92***	0.92***
Adj R-square	24%	22%	20%	18%

We have estimated models with all the time dummies; however, time dummies are found statically significant only for the years 2011. Here we present only the best-fitted models. The variable description is reported in Table 7

*, **, and *** represent significant at respectively 10%, 5% and 1% level

is not. The result may be justified by the results from Table 18. It shows that in 2011, the dip in TES was experienced by both public and private sector banks. On the other hand, in 2017, the dip in TES was observed only for public sector banks. This result corroborates the statement published in the Financial Services Annual Report of *Ministry of Finance, the Government of India (2021–2022)*. It says “As per Reserve Bank of India (RBI) data, aggregate gross advances of Public Sector Banks (PSBs) increased from Rs.18,19,074 crores as on 31.3.2008 to Rs.52,15,920/ crores as on 31.3.2014. As per RBI inputs, aggressive lending practices, during this period along with willful default/loan frauds/ corruption in some cases, economic slowdown, etc. were observed to be primary reasons for the spurt in the stressed assets.” As an aftermath of the global financial crisis (GFC), competition in lending has led to highly stressed assets followed by the NPA crisis.

Our estimated regression results of *panel estimations I, II, II.A, III, and IV* are reported in the Tables 11, 12, 13, 14, and 15 respectively. Our results in this regard for each of these alternative models show similar kind of findings. Importantly,

for all our estimated models, we find negative and significant relationship with banks TES and their priority sector lending exposure. It might be an important finding as priority sector loan gets special emphasis due to their social benefit and overall contribution to economic growth. Interestingly, the negative and significant coefficient values for *maturity* in models relating to TE 3 and TE 4 reveal that the banks manage credit risk more efficiently for short-term loan portfolios. As the tenure of loans increases, banks' TES reduces. Another important and interesting finding is the positive and significant coefficient values for a *secured loan*, especially for TE 1, where the desired output is only *advances*. So, we can conclude that a

Table 11 Panel estimation—I: impact of bank specific factors on TES. Source: Authors' own estimation

	TE1		TE2		TE3		TE4	
	FE	RE	FE	RE	FE	RE	FE	RE
Maturity	− 0.0004	− 0.0006	− 0.0004	− 0.0008**	− 0.00005	− 0.0008**	0.0002	− 0.0005*
SecLoan	0.0013*	0.0012**	0.0007	0.0006	− 0.0002	0.0001	− 0.0000109	0.0002
PriSecL	− 0.0017*	− 0.0017**	− 0.0002	− 0.0007	− 0.0011	− 0.0014**	− 0.0008	− 0.001*
CD	0.004***	0.0039***	0.005***	0.0043***	0.0036***	0.003***	0.003***	0.0025***
Provision	0.0098***	0.0094***	0.0035	0.003	0.0010	0.0001	− 0.0005	− 0.0011
TL	− 0.011**	− 0.0097**	− 0.0011	− 0.001	− 0.0085**	− 0.0065***	− 0.007**	− 0.0053*
_cons	0.73***	0.73***	0.59***	0.65***	0.84***	0.89***	0.84***	0.89***

We could not reject the null hypothesis of Hausman test for any of the models. It means there exists no systematic difference between fixed effects and random effects model. Variable description is reported in Table 7

All the models have been tested with time fixed effect along with entity specific fixed effect. However, we have not found significant time fixed effect for Models 2, 3 and 4. For Model 1, with time fixed effect we found CD and Provision are significant

*, **, and *** represent significant at respectively 10%, 5% and 1% level

Table 12 Panel estimation—II: impact of bank specific factors, macroeconomic factor and HHI on TES. Source: Authors' own estimation

	TE 1		TE 2		TE 3		TE 4	
	FE	RE	FE	RE	FE	RE	FE	RE
Maturity	− 0.0004	− 0.0006	− 0.0004	− 0.0009**	− 0.00007	− 0.00078**	0.00014	− 0.00051*
SecLoan	0.0019***	0.0017***	0.0009	0.0008	0.0002	0.0003	0.00023	0.00033
PriSecL	− 0.0015	− 0.0014*	− 0.0006	− 0.0009	− 0.00098	− 0.001	− 0.0006	− 0.0007
CD	0.004***	0.0045***	0.0049***	0.0046***	0.0041***	0.0034***	0.0033***	0.0028***
Provision	0.010***	0.0092***	0.0051*	0.004	0.0026	0.0011	0.0008	− 0.00023
HHI	− 0.98	− 0.76	− 0.12	0.1	− 1.817	− 1.85*	− 1.69	− 1.67
GDP	0.0041	0.0040	0.006*	0.006*	0.0069**	0.0064**	0.005**	0.0052**
_cons	0.58***	0.57***	0.5***	0.55***	0.78***	0.87***	0.82***	0.88***

We could not reject the null hypothesis of Hausman test for any of the models. It means there exists no systematic difference between fixed effects and random effects model. Variable description is reported in Table 7

*, **, and *** represent significant at respectively 10%, 5% and 1% level

Table 13 Panel estimation—II.A: impact of bank specific factor and HHI on TES. Source: Authors' own estimation

	TE 1		TE 2		TE 3		TE 4	
	FE	RE	FE	RE	FE	RE	FE	RE
Maturity	0.00003	0.0002	0.0001	0.00005	0.0004	− 0.00005	0.0005	0.00009
SecLoan	0.0013*	0.001	0.0005	0.0003	− 0.0002	− 0.0002	− 0.00005	− 0.00003
Provision	0.0029	0.0016	− 0.0027	− 0.0037	− 0.005**	—	− 0.005**	—
						0.0059***		0.0059***
HHI	− 1.97*	− 1.93*	− 0.52	− 0.4673	− 1.8**	− 1.72**	− 1.62**	− 1.54**
_cons	0.99***	1.005***	0.93***	0.96***	1.12***	1.14***	1.091***	1.11***

We could not reject the null hypothesis of Hausman test for any of the models. It means there exists no systematic difference between fixed effects and random effects model. Variable description is reported in Table 7

*, **, and *** represent significant at respectively 10%, 5% and 1% level

Table 14 Panel estimation—III: impact of bank specific factors and interest rate channel on TES. Source: Authors' own estimation

	TE 1		TE 2		TE 3		TE 4	
	FE	RE	FE	RE	FE	RE	FE	RE
Maturity	− 0.0005	− 0.0007	− 0.0005	− 0.0009**	− 0.0001	− 0.0008**	0.00008	− 0.0005*
SecLoan	0.002***	0.0018***	0.0009	0.0008	0.0004	0.0005	0.00044	0.00052
PriSecL	− 0.0019*	− 0.0017**	− 0.00098	− 0.0013*	− 0.0014*	− 0.0014**	− 0.00098	− 0.001*
CD	0.0041***	0.0043***	0.0045***	0.004***	0.004***	0.0032***	0.0032***	0.0027***
Provision	0.0075**	0.0065**	0.0007	− 0.00006	− 0.0012	− 0.0024	− 0.0022	− 0.0032
LR	− 0.013	− 0.015*	− 0.026***	− 0.028***	− 0.014*	− 0.014**	− 0.011*	− 0.011*
_cons	0.67***	0.69***	0.84***	0.92***	0.83***	0.92***	0.83***	0.90***

We could not reject the null hypothesis of Hausman test for any of the models. It means there exists no systematic difference between fixed effects and random effects model. Variable description is reported in Table 7

*, **, and *** represent significant at respectively 10%, 5% and 1% level

Table 15 Panel estimation—IV: impact of bank-specific factors and ownership pattern on TES. Source: Authors' own estimation

	TE 1	TE 2	TE 3	TE 4
	RE	RE	RE	RE
Maturity	0.0001	− 0.0001	− 0.0002	− 0.0001
SecLoan	0.0018***	0.0007	0.0005	0.0005
PriSecL	− 0.003***	− 0.0025***	− 0.0024***	− 0.0019***
Provision	0.005**	0.0004	− 0.0031	− 0.0037*
D	0.042**	0.041***	0.022**	0.021**
_cons	0.86***	0.95***	1.011***	0.99***

We have estimated only the random effects models here. Variable description is reported in Table 7

*, **, and *** represent significant at respectively 10%, 5% and 1% level

bank's technical efficiency increases as its ratio of *secured loan* to *total loan* increases. Moreover, it is revealed that the lending practices have further chances to improve the *credit to deposit* (CD) ratio, which in turn would improve the banks' efficiency in managing their credit risk.

In addition, we also observe that the coefficient of the HHI is negative and significant for the models presented in the Tables 12 and 13. (Table 13 exhibits the results without GDP, presence of which is the cause of insignificant estimated coefficients of HHI due to high positive correlation between the two as shown in the Table 5). Given that HHI is a representation of market concentration, we can conclude that banks' operational efficiency to extend *advances* (good loan) and recovery fall with the increase in market concentration (i.e., reduction of market competition). Moreover, Table 12 also reveals that a bank's technical efficiency in recovery of stressed assets increases with GDP (in TE 2 and TE 3). At the same time, economic boom can also lead to more advances, and better recovery with a simultaneous fall in accumulation of bad loans. This finding is significant to understand the cyclicity in credit risk of Indian commercial banks happens with the macroeconomic boom and bust. It implies that, besides banks' own performance parameters, macroeconomic fundamentals also play noteworthy role in determining operational efficiency.

Table 14 presents the impact of interest rate channel in determining TES of banks in managing their credit risk. To avoid the problem of multicollinearity (as LR has high and significant negative correlation with HHI as shown in the Table 5), the effect of LR has been tested separately (i.e., in absence of the HHI and GDP). We observe it to have a negative and significant coefficient for each of the models. It implies that the overall operational efficiency of the banks to manage their credit risk portfolio improves with a reduction in the lending rate. This is due to the fact that a decrease in the cost of funds would have some favorable effect on the demand for loans. At the same time, it is likely that the recovery of stressed assets improves with a decrease in the cost of funds. Table 15 reveals that private sector banks are overall better in handling both *advances* and *recovery* compared to the public sector banks. This corroborates our preliminary results and the findings from cross-sectional regressions.

Conclusion

Our research delves into the operational efficiency of Indian banks between 2009–2010 and 2017–2018, with a focus on their ability to manage credit risk. Our analyses consider *advances* and *recovery* of stressed assets as desirable outputs, while *Non-Performing Assets* (NPA) and *slippage* are deemed undesirable. This is a crucial area of study, given the fact that the Indian banking sector is stressed with high GNPA and low recovery rate of stressed assets. To estimate four sets of inter-linked operational efficiency scores, we utilize the data envelopment analysis (DEA) method and discover that no previous study has measured the technical efficiency score (TES) of banks in cash recovery of stressed assets. By using TES as a yardstick, we examine bank-specific factors, competition within the banking industry,

overall macroeconomic growth, and the impact of interest rates to understand their effect on bank performance.

It has been observed that both Indian public and private sector banks experience an increase in GNPA over time. However, the rate of increment is much faster for public sector banks. On average, public sector banks only recover 9% of GNPA, while for private sector banks it is around 17% during the study period. The recovery of stressed assets has sharply declined for public sector banks since 2015, while the declining trend for private sector banks is not so sharp and has even slightly improved in the last year. TESs reveal a widening gap between public and private sector banks in terms of advances and/or recovery of stressed assets. It is interesting to note that, on average, a public (private) sector bank can expand its advances and recovery while reducing slippage by 4.7% (2.0%) without increasing its input usage.

Our second stage econometric estimations have led us to conclude that all explored bank-specific variables show a negative and significant relationship with the TES of banks and priority sector lending norms. However, we must consider the social benefits or positive externalities generated through the societal obligatory role that banks play through such norms. Priority sector lending is given more emphasis by regulators to achieve a more inclusive growth path for the economy. Nonetheless, banks are found to be more efficient in credit risk management for shorter-term loan portfolios, and almost all analytical models suggest that the efficiency of a bank providing advances increases with an increase in the share of secured loans to total loans. This requires policy attention to encourage banks to mandate collateral that can be easily convertible to liquidity in case of probable default. In other words, financial assets that earn fixed returns might enhance the feasibility of cash recovery, especially for large-value loans that might lead to extreme value losses, if defaulted. This might be a better alternative and viable solution for loans with longer maturity.

Our findings also suggest that banks' operational efficiency in extending advances and recovering stressed assets decreases with the increase in market concentration of lending. Therefore, banks must be more vigilant in handling high-value accounts to manage their credit risk better. Public sector banks have more scope to enhance their credit risk management to boost their technical efficiency in lending, while private sector banks are doing better in this regard. Moreover, we have observed that the efficiency in the recovery of bad loans is low during the economic downturn. This finding is due to the existence of pro-cyclicality in the financial markets, which is highly linked to the economic fundamentals of a country.

Appendix

See Tables 16, 17, 18.

Table 16 Individual bank-wise average TE score during 2009–2010 to 2017–2018. Source: Authors' own estimation

	Average TES of the public sector banks (in %)			
	TE 1	TE 2	TE 3	TE 4
Allahabad Bank	87.07	88.18	90.00	90.45
Andhra Bank	96.72	96.28	96.94	97.33
Bank of Baroda	95.75	97.79	98.65	99.19
Bank of India	91.05	93.05	96.56	94.96
Bank of Maharashtra	86.04	88.07	90.46	91.44
Canara Bank	98.34	95.98	98.92	99.02
Central Bank of India	80.02	82.91	90.95	92.64
Corporation Bank	99.88	99.56	99.70	99.55
Dena Bank	83.22	84.40	87.90	88.62
IDBI Bank	100.00	100.00	100.00	100.00
Indian Bank	94.03	92.37	93.84	94.46
Indian Overseas Bank	89.74	90.56	97.65	97.67
Oriental Bank of Commerce	90.54	92.23	97.17	98.06
Punjab and Sind Bank	93.91	93.33	94.46	94.86
Punjab National Bank	98.70	96.06	98.42	98.25
Syndicate Bank	92.14	92.35	95.95	96.45
UCO Bank	96.97	94.09	98.48	98.26
Union Bank of India	94.17	94.51	96.53	96.88
United Bank of India	76.16	77.17	85.31	87.31
Vijaya Bank	89.27	89.12	93.19	92.78
State Bank of India & its associates	100.00	100.00	100.00	100.00
	Average TES of the private sector banks (in %)			
	TE 1	TE 2	TE 3	TE 4
Catholic Syrian Bank Ltd	93.78	95.32	97.83	97.33
City Union Bank Ltd	97.50	95.83	98.36	98.47
Federal Bank Ltd	91.93	92.73	98.86	99.18
Jammu & Kashmir Bank Ltd	83.37	90.37	92.75	94.27
Karnataka Bank Ltd	100.00	100.00	100.00	100.00
Karur Vysya Bank Ltd	92.70	92.69	94.09	95.57
Lakshmi Vilas Bank Ltd	96.33	98.53	100.00	100.00
Nainital Bank Ltd	100.00	97.91	100.00	100.00
Ratnakar Bank Ltd	100.00	100.00	100.00	100.00
South Indian bank Ltd	95.34	93.53	95.50	96.09
Tamilnad Mercantile Bank Ltd	93.73	92.05	95.06	95.31
Development Credit Bank Ltd	99.96	97.92	98.84	98.73
HDFC Bank Ltd	100.00	100.00	100.00	100.00
ICICI Bank Ltd	100.00	100.00	100.00	100.00
IndusInd Bank Ltd	97.56	94.86	96.24	96.78
Kotak Mahindra Bank Ltd	95.73	97.39	97.99	98.21
Axis Bank Ltd	97.31	94.28	95.02	95.76
Yes Bank Ltd	100.00	100.00	100.00	100.00

Table 17 Individual bank-wise scope of further improvement. Source: Authors' estimation

	Scope of improvement (in %) for the public sector banks			
	TE 1	TE 2	TE 3	TE 4
Allahabad Bank	14.85	13.41	11.11	10.56
Andhra Bank	3.40	3.87	3.15	2.74
Bank of Baroda	4.44	2.26	1.37	0.82
Bank of India	9.83	7.47	3.56	5.31
Bank of Maharashtra	16.22	13.55	10.55	9.36
Canara Bank	1.69	4.19	1.09	0.99
Central Bank of India	24.97	20.61	9.95	7.94
Corporation Bank	0.12	0.44	0.30	0.45
Dena Bank	20.17	18.48	13.77	12.85
IDBI Bank	0.00	0.00	0.00	0.00
Indian Bank	6.35	8.26	6.56	5.86
Indian Overseas Bank	11.43	10.42	2.40	2.38
Oriental Bank of Commerce	10.45	8.42	2.91	1.97
Punjab and Sind Bank	6.48	7.15	5.86	5.42
Punjab National Bank	1.31	4.11	1.61	1.78
Syndicate Bank	8.53	8.28	4.22	3.68
UCO Bank	3.13	6.28	1.55	1.77
Union Bank of India	6.19	5.81	3.59	3.22
United Bank of India	31.31	29.59	17.21	14.53
Vijaya Bank	12.02	12.21	7.31	7.79
State Bank of India and its associates	0.00	0.00	0.00	0.00
	Scope of improvement (in %) for the private sector banks			
	TE 1	TE 2	TE 3	TE 4
Catholic Syrian Bank Ltd	6.63	4.91	2.22	2.74
City Union Bank Ltd	2.56	4.35	1.67	1.56
Federal Bank Ltd	8.78	7.84	1.15	0.83
Jammu & Kashmir Bank Ltd	19.95	10.66	7.81	6.07
Karnataka Bank Ltd	0.00	0.00	0.00	0.00
Karur Vysya Bank Ltd	7.88	7.89	6.28	4.63
Lakshmi Vilas Bank Ltd	3.81	1.50	0.00	0.00
Nainital Bank Ltd	0.00	2.13	0.00	0.00
Ratnakar Bank Ltd	0.00	0.00	0.00	0.00
South Indian bank Ltd	4.88	6.92	4.72	4.07
Tamilnad Mercantile Bank Ltd	6.69	8.63	5.20	4.92
Development Credit Bank Ltd	0.04	2.13	1.18	1.28
HDFC Bank Ltd	0.00	0.00	0.00	0.00
ICICI Bank Ltd	0.00	0.00	0.00	0.00
IndusInd Bank Ltd	2.50	5.41	3.91	3.33
Kotak Mahindra Bank Ltd	4.46	2.68	2.06	1.82
Axis Bank Ltd	2.77	6.07	5.24	4.43
Yes Bank Ltd	0.00	0.00	0.00	0.00

Table 18 Ownership-wise performance of Indian commercial banks. Source: Authors' estimation

TES (in %)								
Year	TE1		TE2		TE3		TE4	
	Public	Private	Public	Private	Public	Private	Public	Private
2010	88.87	92.54	90.99	94.00	93.82	97.42	94.23	97.51
2011	88.02	92.87	90.09	93.22	94.01	94.36	94.52	95.25
2012	90.55	95.41	91.16	95.48	96.41	98.49	96.69	98.82
2013	93.58	97.96	93.79	95.91	97.32	97.77	97.18	97.96
2014	94.99	98.39	92.50	98.02	95.51	99.44	95.96	99.51
2015	95.90	97.50	95.49	98.27	97.78	99.12	97.84	99.35
2016	93.19	97.07	93.21	96.31	97.66	98.03	97.84	98.18
2017	92.51	97.46	91.61	97.74	92.67	98.14	93.46	98.69
2018	92.16	96.64	90.22	97.45	91.81	97.33	92.43	97.47

Acknowledgements We are immensely thankful to all the reviewers for their insightful comments. We are thankful to the Reserve Bank of India (RBI) for granting us the opportunity to carry out this work under the “Scholarship Scheme for Faculty Members from Academic Institutions, 2020”. We also gratefully acknowledge the constructive comments and suggestions from the scholars of the RBI, which immensely help us revise the paper. However, we are solely responsible for the views expressed here and neither the RBI nor the institutes we are affiliated with are accountable for those.

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